

Java Printing Variables

Combining Data and Output for Dynamic Programs

1. The Power of the Plus (+) Symbol

In Java, the + symbol is smart. Its behavior changes based on what you are giving it.

1. CONCATENATION (JOINING)

When used with **Text (Strings)**, it glues them together.

```
"Hello " + "John"
```

→ Output: "Hello John"

2. ADDITION (MATH)

When used with **Numbers**, it calculates the sum.

```
5 + 6
```

→ Output: 11

2. Mixing Text and Numbers

Java reads code from **Left to Right**. This is critical when you mix text and math in the same print statement.

The "105" Mistake:

```
System.out.println("Result: " + 10 + 5);
```

1. Step 1: "Result: " + 10 → "Result: 10"

2. Step 2: "Result: 10" + 5 → "Result: 105"

Outcome: Result: 105

3. Fixing It with Parentheses ()

If you want math to happen first, wrap it in parentheses. Java always solves the inside of parentheses before anything else.

```
// Correct approach:  
System.out.println("Result: " + (10 + 5));  
Outcome: Result: 15
```

4. Practical Examples

Example: User Greeting

```
String name = "Alice";  
System.out.println("Hello " + name);
```

Example: Full Name Concatenation

```
String fName = "John";  
String lName = "Doe";  
System.out.println(fName + " " + lName);
```

5. Common Pitfalls

- **✗ No Plus:** `System.out.println("Name: " name);` ← Error: Java doesn't know how to join them.
- **✗ Math Ordering:** `"Score: " + x * y` actually works because `*` is higher priority than `+`, but adding `"Score: " + x + y` will fail to add them together.

6. Summary Checklist

Join Mode (+)

At least one side is a String. Results in text.

Parentheses

Forces math to happen before joining text.

Math Mode (+)

Both sides are numbers. Results in a sum.

Sequence

Java evaluates left to right (mostly).

Remember: Semicolon at the end of every Print statement! ;

Codehive